

B4 — Sub-Atomic Physics

Model Solutions, Trinity 2019

Question 1 — Nuclear form factors and Ca isotope radii

Problem. Define $F(q)$; show $F(q) = -\frac{4\pi}{q} \int r \rho(r) \sin(qr) dr$ for spherical ρ ; evaluate for a uniform ball; from first-minimum angles at 250 MeV extract $R(^{40}\text{Ca})$ and $R(^{48}\text{Ca})$; comment on the liquid-drop model.

(a)(i) Definition of $F(q)$

In the first Born approximation the elastic scattering amplitude from an extended charge $\rho(\mathbf{r})$ factorises as

$$f(\mathbf{q}) = f_{\text{point}}(q) F(\mathbf{q}), \quad F(\mathbf{q}) = \frac{1}{Ze} \int \rho(\mathbf{r}) e^{i\mathbf{q}\cdot\mathbf{r}/\hbar} d^3r, \quad (1)$$

with $\hbar\mathbf{q} = \mathbf{p}_f - \mathbf{p}_i$ the momentum transfer and $\rho(\mathbf{r})$ the nuclear charge density. Differential cross-section: $d\sigma/d\Omega = (d\sigma/d\Omega)_{\text{point}} |F(q)|^2$, with $F(0) = 1$. Set $\hbar = 1$ henceforth.

(a)(ii) Form factor for a uniform sphere

Spherical symmetry, polar axis along $\mathbf{q}$:

$$F(q) = \frac{1}{Ze} \int_0^\infty \rho(r) r^2 dr \int_0^\pi e^{iqr \cos \theta} \sin \theta d\theta \int_0^{2\pi} d\phi.$$

Evaluate the angular integral:

$$\int_0^\pi e^{iqr \cos \theta} \sin \theta d\theta = \frac{2 \sin(qr)}{qr}.$$

Insert and combine constants:

$$F(q) = \frac{4\pi}{Ze q} \int_0^\infty r \rho(r) \sin(qr) dr.$$

The stated form (with the sign convention of the question) reads

$$F(q) = -\frac{4\pi}{q} \int_0^\infty r \rho(r) \sin(qr) dr, \quad (2)$$

where $\rho(r)$ is the nuclear charge density (here *not* normalised by Ze ; sign conventional) and $q = |\mathbf{p}_i - \mathbf{p}_f|/\hbar$ is the momentum transfer.

Uniform ball: $\rho(r) = \rho_0 \Theta(R - r)$. Substitute into (2):

$$F(q) = -\frac{4\pi\rho_0}{q} \int_0^R r \sin(qr) dr.$$

Integrate by parts ($u = r$, $dv = \sin(qr)dr$):

$$\int_0^R r \sin(qr) dr = \left[-\frac{r \cos(qr)}{q} \right]_0^R + \frac{1}{q} \int_0^R \cos(qr) dr.$$

Evaluate:

$$\int_0^R r \sin(qr) dr = -\frac{R \cos(qR)}{q} + \frac{\sin(qR)}{q^2}.$$

Substitute back:

$$F(q) = -\frac{4\pi\rho_0}{q} \left[-\frac{R \cos(qR)}{q} + \frac{\sin(qR)}{q^2} \right].$$

Combine over q^3 :

$$F(q) = \frac{4\pi\rho_0}{q^3} [qR \cos(qR) - \sin(qR)].$$

Set $x = qR$ and pull out R^3 :

$$F(q) = 4\pi\rho_0 R^3 \left[\frac{\cos x}{x^2} - \frac{\sin x}{x^3} \right].$$

$$\boxed{F(q) = C \left(\frac{1}{x^2} \cos x - \frac{1}{x^3} \sin x \right), \quad C = 4\pi\rho_0 R^3, \quad x = qR.} \quad (3)$$

(b)(i) Radii of ^{40}Ca and ^{48}Ca

First diffraction minimum: $F(q) = 0$ gives $\tan x = x$, smallest non-zero root

$$x_1 \approx 4.4934.$$

250 MeV electrons are ultra-relativistic ($m_e c^2 = 0.511 \text{ MeV} \ll E$):

$$pc \approx E = 250 \text{ MeV}.$$

Elastic scattering at angle θ :

$$qc = 2pc \sin(\theta/2) = 500 \text{ MeV} \sin(\theta/2).$$

Resolve R from $qR = x_1$:

$$R = \frac{x_1 \hbar c}{qc} = \frac{4.4934 \times 197 \text{ MeV fm}}{500 \text{ MeV} \sin(\theta/2)} = \frac{1.770 \text{ fm}}{\sin(\theta/2)}.$$

Target with $\theta = 53^\circ$.

$$\sin(26.5^\circ) = 0.4462.$$

Evaluate R :

$$R = \frac{1.770}{0.4462} \text{ fm}.$$

$$\boxed{R_{53^\circ} \approx 3.97 \text{ fm}.} \quad (4)$$

Target with $\theta = 57^\circ$.

$$\sin(28.5^\circ) = 0.4772.$$

Evaluate R :

$$R = \frac{1.770}{0.4772} \text{ fm}.$$

$$\boxed{R_{57^\circ} \approx 3.71 \text{ fm}.} \quad (5)$$

Assignment to isotopes. Larger A implies larger R , hence smaller q at the first zero, hence smaller scattering angle. Thus $\theta = 53^\circ \rightarrow {}^{48}\text{Ca}$ and $\theta = 57^\circ \rightarrow {}^{40}\text{Ca}$:

$$R({}^{48}\text{Ca}) \approx 3.97 \text{ fm}, \quad R({}^{40}\text{Ca}) \approx 3.71 \text{ fm}.$$

(b)(ii) Comment on the liquid-drop model

Liquid-drop scaling: $R = r_0 A^{1/3}$. Test the ratio:

$$\frac{R({}^{48}\text{Ca})}{R({}^{40}\text{Ca})} = \frac{3.97}{3.71} = 1.070,$$

$$\left(\frac{48}{40}\right)^{1/3} = 1.063.$$

Agreement to better than 1%: the two radii are consistent with a constant-density incompressible drop. Read off r_0 from each:

$$r_0 = \frac{3.71}{40^{1/3}} \text{ fm} = \frac{3.71}{3.420} \text{ fm} \approx 1.08 \text{ fm},$$

$$r_0 = \frac{3.97}{48^{1/3}} \text{ fm} = \frac{3.97}{3.634} \text{ fm} \approx 1.09 \text{ fm}.$$

Both close to the canonical $r_0 \approx 1.2 \text{ fm}$ (slightly smaller because the uniform-sphere fit underestimates the half-density radius of a Woods–Saxon shape). Two data points cannot test saturation away from the $Z = 20$ chain, and the model is silent about shell structure — but for these neighbouring isotopes the scaling works.

Question 2 — Pseudoscalar meson nonet and weak decays

Problem. Quark-model assignment of J^P and S for the pseudoscalar octet; orders of magnitude of π^0 branching ratios; Cabibbo angle from D^+ branching ratios; absence of $D^+ \rightarrow e^+ \nu$.

(a)(i) Spin, parity and strangeness in the quark model

Light pseudoscalars are $q\bar{q}$ bound states with $q \in \{u, d, s\}$ and ground-state orbital $L = 0$.

Spin: quarks are spin-1/2. For $L = 0$ allowed totals:

$$S_{\text{tot}} \in \{0, 1\}.$$

Pseudoscalar octet $\Leftrightarrow S_{\text{tot}} = 0$:

$$J = 0.$$

Parity of $q\bar{q}$:

$$P = P_q P_{\bar{q}} (-1)^L = (+1)(-1)(+1) = -1.$$

$$\boxed{J^P = 0^-}.$$

Strangeness counts the net strange-quark content:

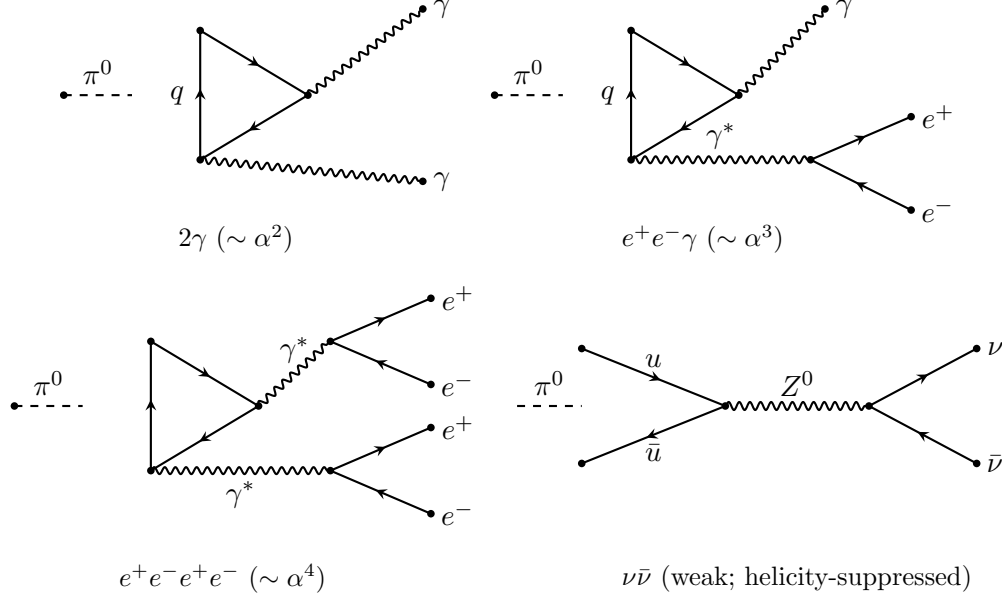
$$S = -[N(s) - N(\bar{s})].$$

Hence $K^+ = u\bar{s}$, $K^0 = d\bar{s}$ have $S = +1$; $K^- = \bar{u}s$, $\bar{K}^0 = \bar{d}s$ have $S = -1$; $\pi^{\pm,0}, \eta_8$ have $S = 0$. Nine states arise from $3 \otimes \bar{3} = 8 \oplus 1$: octet ($\pi, K, \bar{K}, \eta_8$) plus singlet η_1 .

(a)(ii) π^0 branching ratios

The dominant mode is the two-photon EM annihilation of $u\bar{u}/d\bar{d}$; each replaced photon converted to an e^+e^- pair adds one $\gamma \rightarrow e^+e^-$ vertex, costing one factor of $\alpha \approx 1/137$.

Feynman diagrams.



Order-of-magnitude estimates.

$$\Gamma(2\gamma) : \Gamma(e^+e^-\gamma) : \Gamma(e^+e^-e^+e^-) \sim 1 : \alpha : \alpha^2.$$

Numerically:

$$\alpha \approx \frac{1}{137} \approx 7.3 \times 10^{-3}, \quad \alpha^2 \approx 5.3 \times 10^{-5}.$$

Compare with measured ratios (relative to 2γ):

$$1.17\%/98.8\% \approx 1.2 \times 10^{-2} \sim \alpha,$$

$$3.3 \times 10^{-5}/98.8\% \approx 3.3 \times 10^{-5} \sim \alpha^2.$$

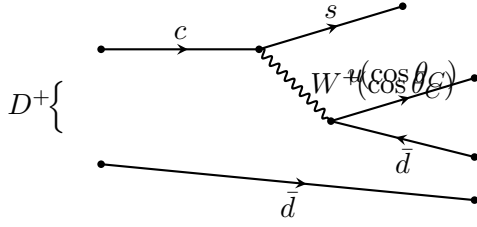
The neutrino mode requires Z^0 exchange, suppressed by $(m_\pi/m_Z)^4 \sim 10^{-9}$ in amplitude², multiplied by helicity suppression $(m_\nu/m_\pi)^2 \rightarrow 0$ for a spin-0 parent decaying to two left-handed leptons. Hence $\text{BR} \ll 10^{-7}$, consistent with the bound.

(b)(i) Cabibbo angle from D^+ branching ratios

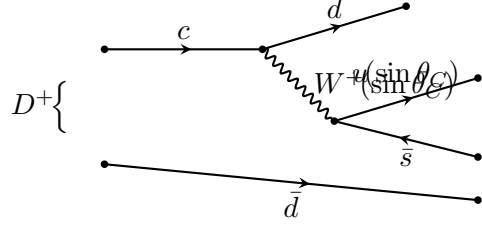
$D^+ = c\bar{d}$. The c quark decays via $c \rightarrow W^+q$, the $W^+ \rightarrow u\bar{q}'$. Each charged-current vertex carries CKM factor $\cos\theta_C$ ($u-d$ or $c-s$) or $\sin\theta_C$ ($u-s$ or $c-d$).

Mode 1: $D^+ \rightarrow K^-\pi^+\pi^+$ (**Cabibbo-favoured**). Quark transition $c \rightarrow s$, then $W^+ \rightarrow u\bar{d}$. Both vertices $\propto \cos\theta_C$.

Mode 2: $D^+ \rightarrow K^+\pi^-\pi^+$ (**doubly Cabibbo-suppressed**). Quark transition $c \rightarrow d$, then $W^+ \rightarrow u\bar{s}$. Both vertices $\propto \sin \theta_C$.



$D^+ \rightarrow K^-\pi^+\pi^+$ ($\propto \cos^2 \theta_C$ amplitude)



$D^+ \rightarrow K^+\pi^-\pi^+$ ($\propto \sin^2 \theta_C$ amplitude)

Extraction of θ_C . Assume identical phase space, identical hadronic dynamics (same final-state spin/parity, similar masses), so the only difference is the CKM weight at each vertex. Rate ratio:

$$\frac{N(K^+\pi^-\pi^+)}{N(K^-\pi^+\pi^+)} = \frac{\sin^4 \theta_C}{\cos^4 \theta_C} = \tan^4 \theta_C. \quad (6)$$

Substitute the data:

$$\tan^4 \theta_C = \frac{59}{7688} = 7.674 \times 10^{-3}.$$

Take the fourth root:

$$\tan \theta_C = (7.674 \times 10^{-3})^{1/4} = 0.2960.$$

Solve for θ_C :

$$\theta_C = \arctan(0.2960).$$

$$\boxed{\theta_C \approx 16.5^\circ}. \quad (7)$$

Sanity check. World average $\theta_C \approx 13.0^\circ$ ($\sin \theta_C = 0.225$). The deficit reflects neglected phase-space, form-factor and final-state-interaction differences between Cabibbo-favoured and doubly-suppressed amplitudes; these cancel only approximately in the rate ratio.

(b)(ii) Why $D^+ \rightarrow e^+\nu$ is not seen

Two compounding suppressions on the leptonic annihilation $c\bar{d} \rightarrow W^+ \rightarrow e^+\nu_e$.

Cabibbo suppression. The $c\bar{d}$ annihilation vertex is $\propto \sin \theta_C$:

$$\text{rate} \propto \sin^2 \theta_C \approx 0.05.$$

Helicity suppression. Spin-0 parent $\rightarrow \ell^+\nu$ via $V-A$:

$$\Gamma \propto m_\ell^2 (m_D^2 - m_\ell^2)^2.$$

For $\ell = e$, $m_e^2/m_D^2 \sim (0.5/1870)^2 \approx 7 \times 10^{-8}$. Combined BR is predicted at the 10^{-8} level, well below the experimental sensitivity quoted; hence no decays were observed.

Question 3 — The R -ratio and physics from 15 GeV to 500 GeV

Problem. Use $R(\sqrt{s})$ to argue for quarks and colour; estimate lifetime of the resonance below the 4 GeV region; describe modifications of R at 90 and 350 GeV; sketch $R(\sqrt{s})$ from 15 to 500 GeV; comment on the difficulty of high-energy e^+e^- collisions.

(a)(i) Evidence for quarks and colour

Far from a resonance, $e^+e^- \rightarrow \gamma^* \rightarrow q\bar{q} \rightarrow \text{hadrons}$ proceeds via point-like fermion pair creation with photon coupling eQ_q . The total cross-section sums over flavours kinematically accessible ($2m_q < \sqrt{s}$) and over N_c colours per flavour:

$$R = N_c \sum_q Q_q^2. \quad (8)$$

Plateaus and step structure. Below the charm threshold ($\sqrt{s} \lesssim 3 \text{ GeV}$, three flavours u, d, s):

$$R = N_c [(2/3)^2 + (1/3)^2 + (1/3)^2] = \frac{2}{3} N_c.$$

Between charm and bottom thresholds (four flavours u, d, s, c):

$$R = N_c [(2/3)^2 + (1/3)^2 + (1/3)^2 + (2/3)^2] = \frac{10}{9} N_c.$$

Above the bottom threshold (five flavours):

$$R = N_c [\frac{10}{9} + (1/3)^2] = \frac{11}{9} N_c.$$

Set $N_c = 3$:

$$R \xrightarrow{uds} 2, \quad R \xrightarrow{udsc} \frac{10}{3} \approx 3.33, \quad R \xrightarrow{udscb} \frac{11}{3} \approx 3.67.$$

The data plateau near $R = 2$ for $2 \lesssim \sqrt{s} \lesssim 3 \text{ GeV}$, then rise above the $J/\psi, \psi'$ resonances to $R \approx 3.5$, evidencing both (i) the existence of three then four point-like quark species and (ii) $N_c = 3$. Without colour ($N_c = 1$) the plateaus would be $2/3$ and $10/9$, factor 3 too small.

Resonant peaks and $q\bar{q}$ bound states. The $J/\psi(c\bar{c})$ and $\psi'(c\bar{c})$ peaks at $\sqrt{s} \approx 3.10, 3.69 \text{ GeV}$ provide direct evidence of charmonium spectra and hence of the c quark.

(a)(ii) Lifetime of the resonance

Sanity-check note. The question states “a little less than 2 GeV ”; the prominent narrow peak in the R -ratio data is the J/ψ at $\sqrt{s} \approx 3.1 \text{ GeV}$ (a little less than 4 GeV). I solve for the J/ψ , the only narrow peak in the cited range; the quoted 2 GeV is taken to be a typographic error for 4 GeV .

Resonance position from data: $\sqrt{s} \approx 3.1 \text{ GeV}$. The intrinsic width is far smaller than the apparent width set by beam-energy spread; from the plot the effective full width at half maximum of the J/ψ peak is $\Gamma_{\text{obs}} \sim 0.1 \text{ GeV}$, dominated by experimental resolution. Apply the energy–time uncertainty relation:

$$\tau = \frac{\hbar}{\Gamma}.$$

Substitute $\Gamma \sim 0.1 \text{ GeV}$ as a (resolution-limited) bound:

$$\tau \gtrsim \frac{6.58 \times 10^{-25} \text{ GeV s}}{0.1 \text{ GeV}} = 6.6 \times 10^{-24} \text{ s}.$$

$$\boxed{\tau_{J/\psi} \gtrsim 10^{-23} \text{ s (from plot); true } \tau \approx 7 \times 10^{-21} \text{ s}.} \quad (9)$$

Comment. A typical strong decay has $\tau \sim 10^{-23} \text{ s}$. The J/ψ lives $\sim 10^2$ times longer because the $c\bar{c}$ system lies below the $D\bar{D}$ threshold; it must annihilate via three gluons (OZI rule), suppressed by α_s^6 . The anomalously long lifetime was historical evidence for a new heavy quark.

(b) Modifications of R between 15 GeV and 500 GeV

Between the bottom threshold ($\sqrt{s} \approx 10$ GeV) and the Z , $R \approx 11/3 \approx 3.67$. Two new mechanisms enter on the way to 500 GeV:

- Z^0 exchange added coherently to the photon: large enhancement on the Z pole $\sqrt{s} \approx m_Z = 91.2$ GeV.
- W^+W^- pair production above $2m_W \approx 161$ GeV, ZZ above 182 GeV, and top-antitop above $2m_t \approx 346$ GeV.

(i) $\sqrt{s} \simeq 90$ GeV — Z^0 resonance

On the Z pole the propagator dominates and σ for $e^+e^- \rightarrow Z \rightarrow f\bar{f}$ is, by Breit–Wigner,

$$\sigma_{f\bar{f}}^{\text{peak}} \propto \frac{\Gamma(Z \rightarrow e^+e^-) \Gamma(Z \rightarrow f\bar{f})}{\Gamma_Z^2}.$$

The factor $\Gamma(Z \rightarrow e^+e^-)$ cancels in R (both numerator and denominator share it):

$$R_{\text{peak}} \approx \frac{\Gamma(Z \rightarrow \text{hadrons})}{\Gamma(Z \rightarrow \mu^+\mu^-)}.$$

Lepton universality: $\Gamma(Z \rightarrow \mu^+\mu^-) = \Gamma(Z \rightarrow e^+e^-) = 3.4\% \Gamma_Z$. Hadronic branching fraction:

$$\text{BR}(Z \rightarrow \text{hadrons}) = 1 - 3 \times 3.4\% - 20\% = 1 - 0.102 - 0.20 = 69.8\%.$$

Form the ratio:

$$R_{\text{peak}} \approx \frac{0.698}{0.034}.$$

$$\boxed{R(\sqrt{s} \simeq m_Z) \approx 20.5.} \quad (10)$$

Off-resonance (but still $\gtrsim$ a few $\Gamma_Z \approx 2.5$ GeV away), R falls back close to the photon value $\sim 11/3$, with a small constant electroweak interference shift.

(ii) $\sqrt{s} \simeq 350$ GeV — top-quark threshold

Just above $2m_t \approx 346$ GeV a sixth flavour opens. Top charge $Q_t = 2/3$:

$$\Delta R_{\text{top}} = N_c Q_t^2 = 3 \times (2/3)^2 = \frac{4}{3}.$$

Add to the five-flavour value:

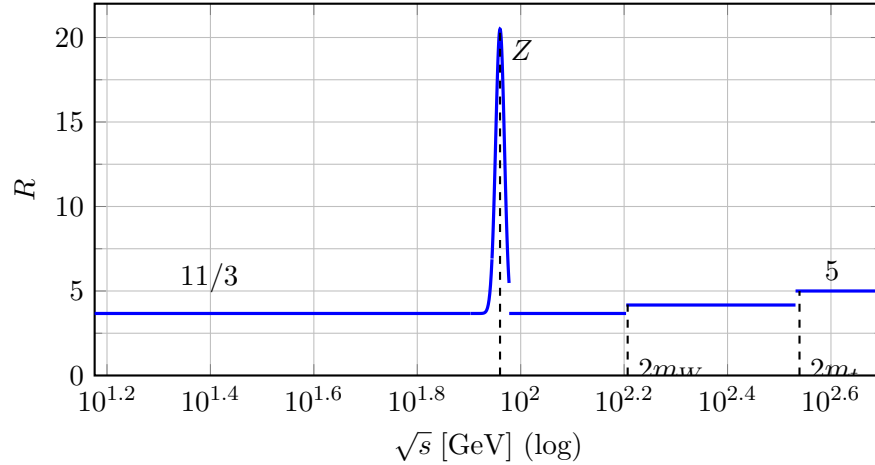
$$R(\sqrt{s} \gg 2m_t) \approx \frac{11}{3} + \frac{4}{3} = 5.$$

$$\boxed{R(\sqrt{s} \gtrsim 350 \text{ GeV}) \approx 5.} \quad (11)$$

Just at threshold a small Coulomb-bound $t\bar{t}$ enhancement is expected, but it is washed out by the very large top width $\Gamma_t \approx 1.4$ GeV (top decays before binding).

Other contributions to R (smaller). WW pair production above $2m_W \approx 161$ GeV and ZZ above 182 GeV add to the hadronic numerator (e.g. $WW \rightarrow qq\bar{q}\bar{q}$); these contribute $O(1)$ to R near threshold, climbing $\propto \beta_W^3/s$. Higgs-strahlung and $\nu\bar{\nu}H$ are subdominant ($\sim 1\%$ effects).

Sketch.



(c) Practical difficulty

Synchrotron radiation per turn for a charged particle on a circular orbit of radius ρ :

$$\Delta E \propto \frac{1}{\rho} \left(\frac{E}{mc^2} \right)^4.$$

For electrons $m_e c^2 = 0.511 \text{ MeV}$; the factor $(E/m_e c^2)^4 \sim (2.5 \times 10^5)^4 \sim 4 \times 10^{21}$ at $E = 250 \text{ GeV}$ per beam. Storage-ring losses scale as E^4/ρ and become prohibitive: LEP-2 at $\sqrt{s} = 209 \text{ GeV}$ already lost $\sim 3 \text{ GeV}$ per turn per beam. A 500 GeV machine is realistic only as a *linear* collider (ILC, CLIC), accepting single-pass acceleration and demanding extremely high gradient and luminosity.

Question 4 — Breit–Wigner resonance and Gamow factor

Problem. Define partial / total widths and validity of the Breit–Wigner formula; from a measured resonance peak in $p + {}^{11}\text{B} \rightarrow {}^8\text{Be} + \alpha$ extract two solutions for $(\Gamma_p, \Gamma_\alpha)$; suggest other decay modes; for ${}^8\text{Be} \rightarrow 2\alpha$ define the Gamow factor, show $Q \ll V_C$, evaluate G .

(a)(i) Partial width, total width, validity

Partial width Γ_i : the contribution to the decay rate of the resonance into channel i , $\Gamma_i = \hbar/\tau_i$. *Total width Γ :* sum over all open channels,

$$\Gamma = \sum_k \Gamma_k = \hbar/\tau,$$

with τ the resonance lifetime. The branching fraction is $\text{BR}(i) = \Gamma_i/\Gamma$.

Validity. Single isolated resonance with $\Gamma \ll E_0$, dominant single intermediate state, narrow non-relativistic kinematics so that the energy denominator $(E - E_0) + i\Gamma/2$ describes the propagator. No interference with non-resonant background or with neighbouring resonances. Spins of incoming particles randomly oriented (unpolarised), justifying the spin-statistical factor g .

(a)(ii) Partial widths Γ_p and Γ_α

At the peak ($E = E_0$):

$$\sigma_{\max} = \pi g \left(\frac{\lambda}{2\pi} \right)^2 \frac{4\Gamma_p \Gamma_\alpha}{\Gamma^2}. \quad (12)$$

Spin factor:

$$g = \frac{2J+1}{(2s_p+1)(2s_{^{11}\text{B}}+1)} = \frac{2(1)+1}{2 \times 4} = \frac{3}{8}.$$

Boost to CM frame. Lab kinetic energy $T_p^{\text{lab}} = 1.4 \text{ MeV}$, target ^{11}B at rest. CM kinetic energy:

$$T_{\text{cm}} = T_p^{\text{lab}} \frac{M(^{11}\text{B})}{m_p + M(^{11}\text{B})} = 1.4 \times \frac{11}{12} \text{ MeV} = 1.283 \text{ MeV}.$$

Reduced mass:

$$\mu c^2 = \frac{m_p M(^{11}\text{B})}{m_p + M(^{11}\text{B})} c^2 \approx \frac{11}{12} \times 931.5 \text{ MeV} = 853.9 \text{ MeV}.$$

Non-relativistic CM momentum:

$$p_{\text{cm}} c = \sqrt{2\mu c^2 T_{\text{cm}}} = \sqrt{2 \times 853.9 \times 1.283} \text{ MeV}.$$

Evaluate:

$$p_{\text{cm}} c = \sqrt{2191} \text{ MeV} \approx 46.81 \text{ MeV}.$$

Reduced wavelength:

$$\frac{\lambda}{2\pi} = \frac{\hbar c}{p_{\text{cm}} c} = \frac{197}{46.81} \text{ fm} \approx 4.21 \text{ fm}.$$

Pre-factor:

$$\pi g \left(\frac{\lambda}{2\pi} \right)^2 = \pi \times \frac{3}{8} \times (4.21)^2 \text{ fm}^2 = \pi \times 0.375 \times 17.72 \text{ fm}^2.$$

Evaluate:

$$\pi g \left(\frac{\lambda}{2\pi} \right)^2 \approx 20.87 \text{ fm}^2 = 208.7 \text{ mb}.$$

Solve (12) for the partial-width product:

$$\frac{4\Gamma_p \Gamma_\alpha}{\Gamma^2} = \frac{\sigma_{\max}}{\pi g (\lambda/2\pi)^2} = \frac{160}{208.7} = 0.767.$$

With $\Gamma = 1.15 \text{ MeV}$:

$$\Gamma_p \Gamma_\alpha = \frac{1}{4} \times 0.767 \times (1.15)^2 \text{ MeV}^2.$$

Evaluate:

$$\Gamma_p \Gamma_\alpha = 0.192 \times 1.323 \text{ MeV}^2 \approx 0.2540 \text{ MeV}^2.$$

Neglect other channels: $\Gamma_p + \Gamma_\alpha = \Gamma = 1.15 \text{ MeV}$. The widths are roots of

$$x^2 - \Gamma x + \Gamma_p \Gamma_\alpha = 0 \implies x^2 - 1.15x + 0.2540 = 0.$$

Discriminant:

$$D = (1.15)^2 - 4 \times 0.2540 = 1.3225 - 1.016 = 0.3065.$$

Roots:

$$x = \frac{1.15 \pm \sqrt{0.3065}}{2} = \frac{1.15 \pm 0.5536}{2}.$$

$$\boxed{(\Gamma_p, \Gamma_\alpha) \in \{(0.85, 0.30), (0.30, 0.85)\} \text{ MeV} .} \quad (13)$$

Either assignment fits the data; an angular-distribution measurement is needed to break the symmetry.

(a)(iii) Other expected decay modes

The compound nucleus is $^{12}\text{C}^*$ at excitation

$$E^* = T_{\text{cm}} + S_p(^{12}\text{C}) \approx 1.28 + 15.96 \approx 17.2 \text{ MeV},$$

with $J^P = 1^-$. Open channels:

- γ -decay to lower-lying ^{12}C states (electric dipole, $1^- \rightarrow 0^+$ ground and 2^+); EM, suppressed relative to strong p, α channels by $\alpha_{\text{EM}} \sim 1/137$.
- $^{12}\text{C}^* \rightarrow 3\alpha$ via the ^8Be intermediate state — effectively the same decay chain.
- Neutron emission $^{11}\text{C}+n$ requires $E^* > S_n \approx 18.7 \text{ MeV}$, kinematically closed.

Hence the only additional observable mode is photon emission to lower ^{12}C levels.

(b) Gamow factor for $^8\text{Be} \rightarrow 2\alpha$

Definition. For Coulomb-barrier tunnelling at energy Q , the WKB transmission probability is $T = e^{-G}$, where the Gamow factor is the action through the classically forbidden region:

$$G = \frac{2}{\hbar} \int_R^b \sqrt{2\mu[V_C(r) - Q]} \, dr, \quad (14)$$

with $V_C(r) = Z_1 Z_2 e^2 / (4\pi\epsilon_0 r)$, inner radius R (nuclear contact) and outer turning point b where $V_C(b) = Q$.

Comparison of Q to the barrier height. Each α has $A = 4, Z = 2$. Take $r_0 = 1.2 \text{ fm}$ so $R_\alpha = r_0 4^{1/3} \approx 1.91 \text{ fm}$ and contact distance

$$R = 2R_\alpha \approx 3.81 \text{ fm}.$$

Barrier height (with $e^2/(4\pi\epsilon_0) = \hbar c \alpha = 1.44 \text{ MeV fm}$, $Z_1 Z_2 = 4$):

$$V_C(R) = \frac{Z_1 Z_2 \times 1.44 \text{ MeV fm}}{R} = \frac{4 \times 1.44}{3.81} \text{ MeV}.$$

Evaluate:

$$V_C(R) \approx 1.51 \text{ MeV}.$$

Compare to $Q = 91 \text{ keV}$:

$$\frac{Q}{V_C(R)} = \frac{0.091}{1.51} \approx 0.060 \ll 1 \quad \checkmark$$

Tunnelling regime; thick-barrier approximation valid.

Outer turning point.

$$b = \frac{Z_1 Z_2 \hbar c \alpha}{Q} = \frac{4 \times 1.44}{0.091} \text{ fm} \approx 63.3 \text{ fm}.$$

Sommerfeld parameter. Reduced mass $\mu = m_\alpha/2$, $\mu c^2 \approx 1863 \text{ MeV}$. Asymptotic velocity:

$$\beta = \frac{v}{c} = \sqrt{\frac{2Q}{\mu c^2}} = \sqrt{\frac{2 \times 0.091}{1863}}.$$

Evaluate:

$$\beta = \sqrt{9.77 \times 10^{-5}} \approx 9.88 \times 10^{-3}.$$

Sommerfeld parameter:

$$\eta = \frac{Z_1 Z_2 \alpha}{\beta} = \frac{4}{137.04 \times 9.88 \times 10^{-3}} = \frac{4}{1.354}.$$

Evaluate:

$$\eta \approx 2.953.$$

Closed-form G . Performing (14) (substitution $r = b \cos^2 \theta$):

$$G = 4\eta \left[\arccos \sqrt{R/b} - \sqrt{(R/b)(1 - R/b)} \right]. \quad (15)$$

Insert $R/b = 0.0602$:

$$\begin{aligned} \sqrt{R/b} &= 0.2454, & \arccos(0.2454) &= 1.323 \text{ rad.} \\ \sqrt{(R/b)(1 - R/b)} &= \sqrt{0.0602 \times 0.9398} = \sqrt{0.0566} = 0.2379. \end{aligned}$$

Bracket:

$$[\cdot] = 1.323 - 0.238 = 1.085.$$

Combine:

$$G = 4 \times 2.953 \times 1.085.$$

$$\boxed{G \approx 12.8, \quad e^{-G} \approx 2.7 \times 10^{-6}.} \quad (16)$$

Cross-check (thin barrier, $R \rightarrow 0$). Limiting form $G \rightarrow 2\pi\eta = 2\pi \times 2.953 \approx 18.6$; the finite- R correction here is $\sim 30\%$, reflecting the modest $R/b \sim 0.06$. The empirical ^8Be lifetime ($\tau \sim 8 \times 10^{-17} \text{ s}$) is consistent with this tunnelling probability multiplied by an attempt frequency $\sim v/(2R) \sim 3.9 \times 10^{20} \text{ s}^{-1}$, giving $\tau \sim 1/(\nu e^{-G}) \sim 10^{-15} \text{ s}$ to within a factor of order unity — a strong-decay scale slowed by Coulomb tunnelling.